



Customer Shopping Behavior Analysis

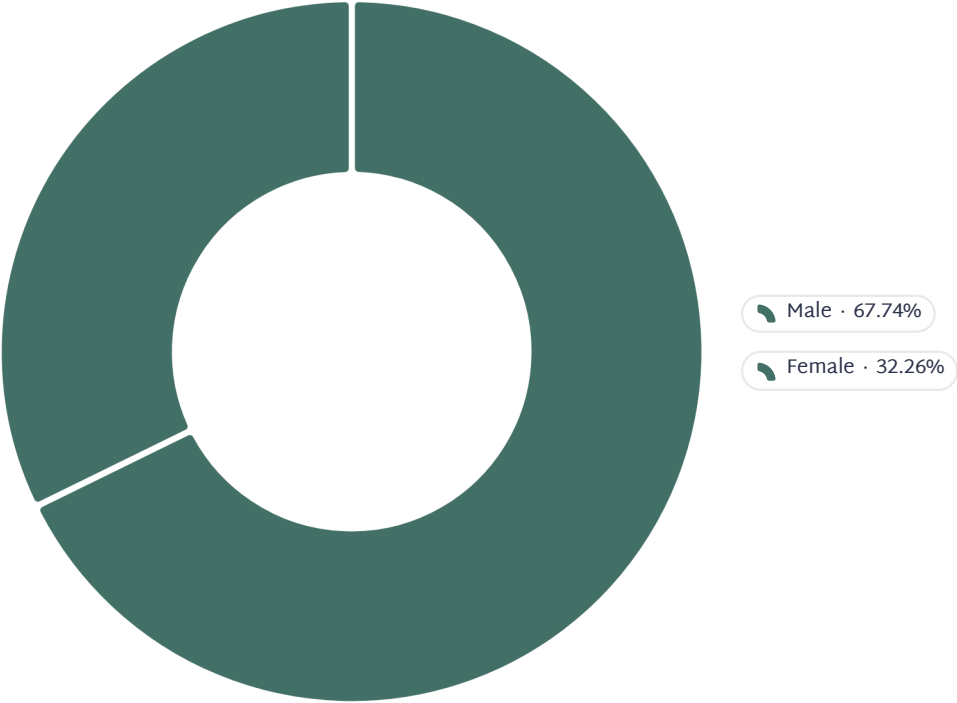
Uncovering patterns across 3,900 transactions to drive smarter retail decisions.



Project Overview & Dataset

- **3,900** customer transactions across 4 product categories
- **18 features** covering demographics, purchase details, and shopping behavior
- Data cleaned in **Python**, analyzed in **SQL**, visualized in **Power BI**

Revenue Distribution by Gender



Revenue Breakdown

Male-dominated revenue signals a clear opportunity to **expand the female customer base** through targeted campaigns and product curation.

Male Customers

\$157,890 — 67.7% of total revenue

Female Customers

\$75,191 — 32.3% of total revenue

Subscription Impact on Revenue

27%

Subscribers

1,053 customers · \$62,645 total revenue
· \$59.49 avg spend

73%

Non-Subscribers

2,847 customers · \$170,436 total
revenue · \$59.87 avg spend

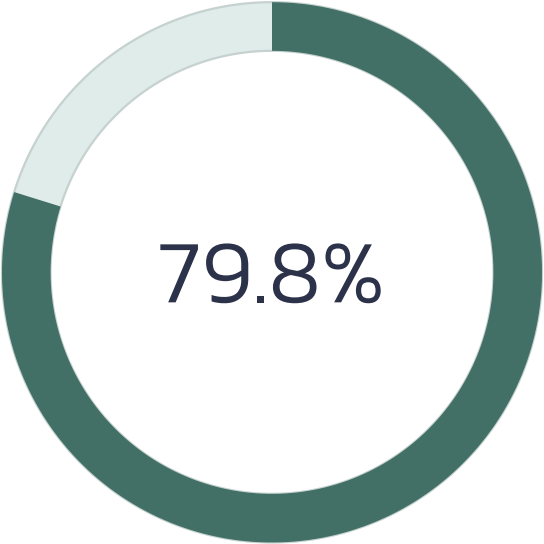
\$0.38

Spend Difference

Nearly identical per-transaction spend
between subscribers and non-
subscribers

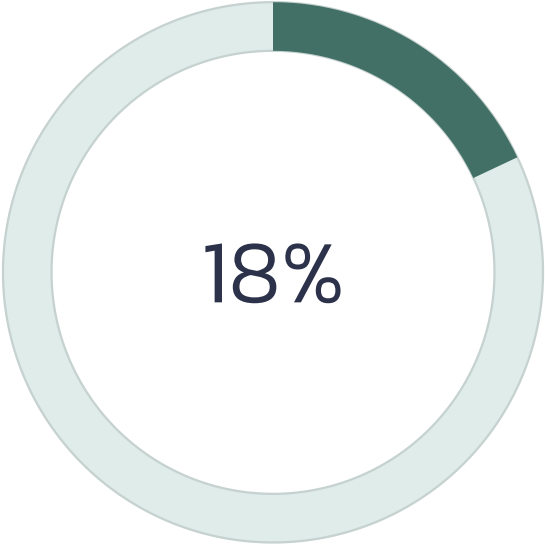
- ❏ Non-subscribers drive **73% of revenue** despite similar per-transaction spend — subscription conversion is a high-leverage growth lever.

Customer Segmentation Results



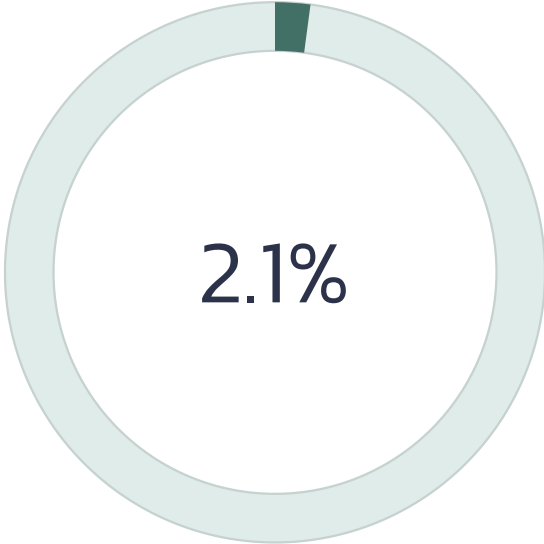
Loyal Customers

3,116 repeat purchasers with strong engagement



Returning Customers

701 with moderate purchase history



New Customers

83 — significant acquisition opportunity

Top Products by Category & Rating

Top Orders by Category

Accessories

Jewelry (171) ·
Sunglasses (161) · Belt (161)

Clothing

Blouse (171) · Pants (171) · Shirt (169)

Highest-Rated Products

 Gloves

Rating: **3.86**

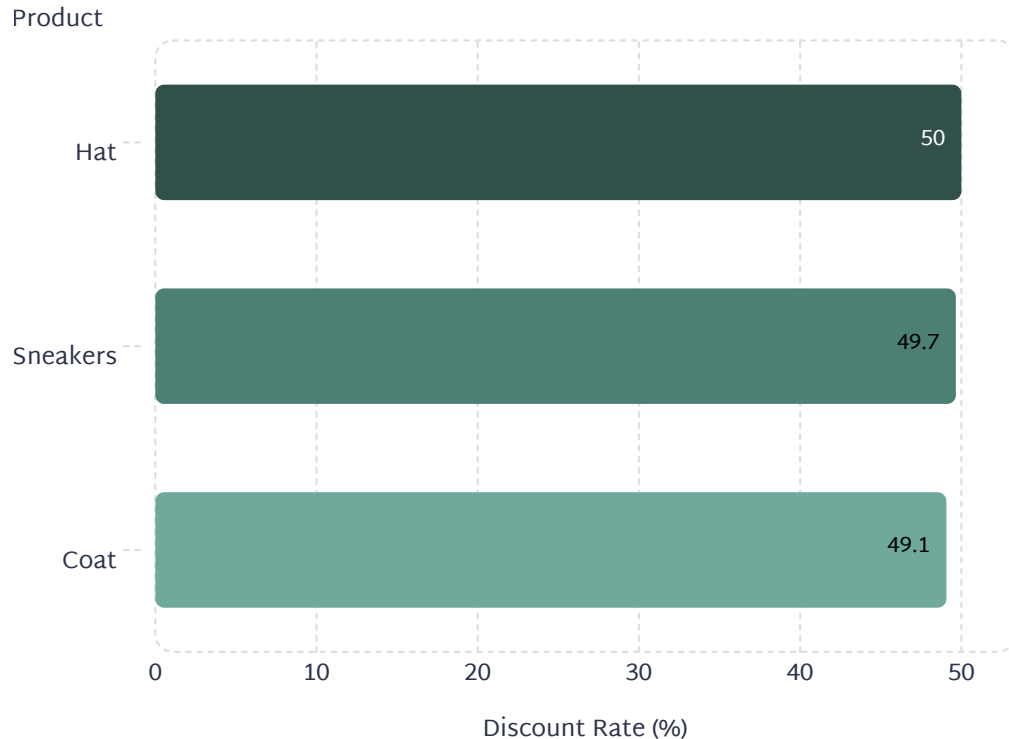
 Sandals

Rating: **3.84**

 Boots

Rating: **3.82**

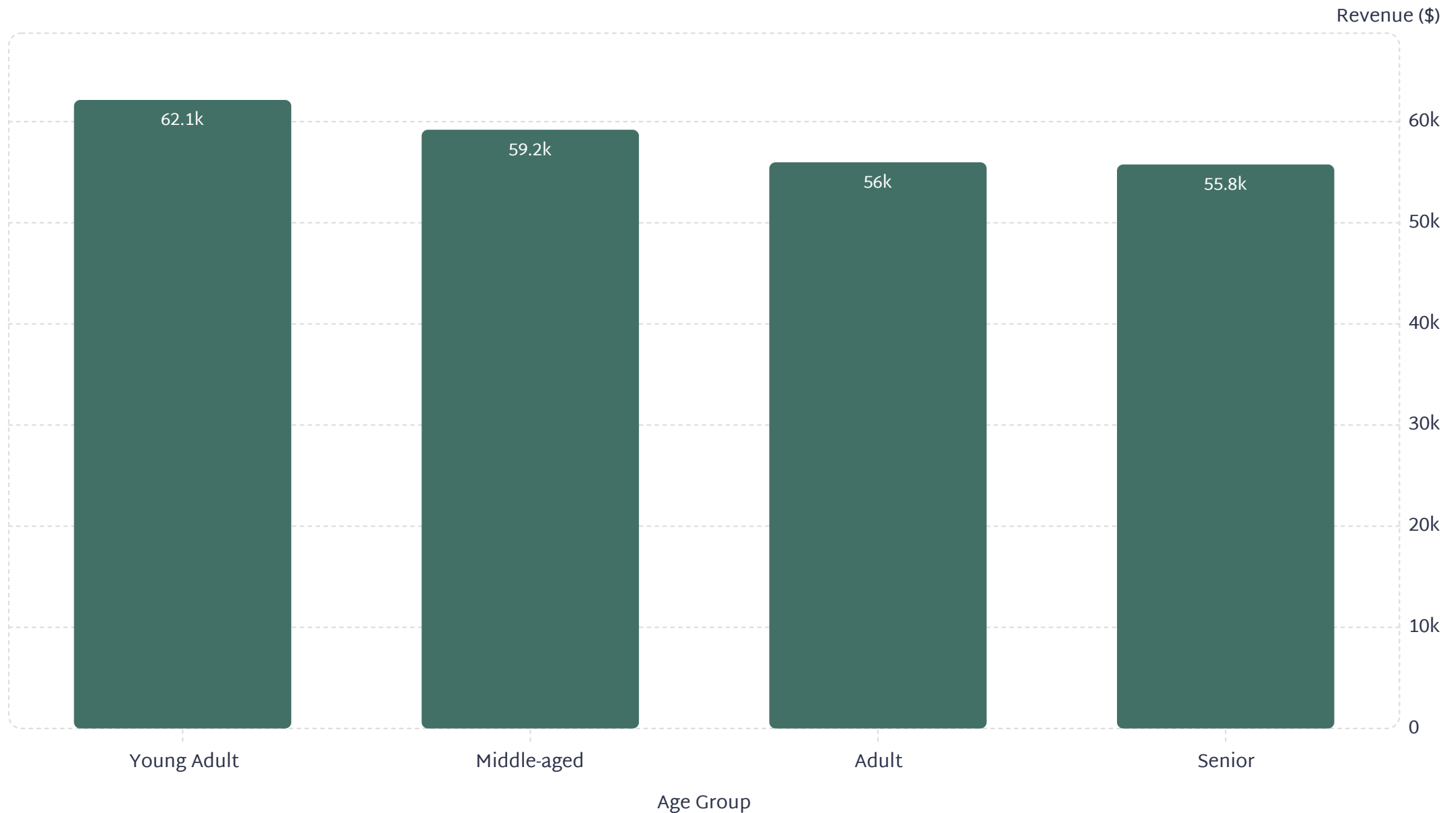
Discount Dependency Analysis



Key Findings

- **Hat** leads with a 50% discount rate — highest in the catalog
- **Sneakers** (49.66%) and **Coat** (49.07%) also heavily discounted
- **839 high-spending discount users** identified — spending above the \$59 average
- Discount strategy drives volume but puts **margins at risk** on key SKUs

Revenue by Age Group



Revenue is **remarkably balanced** across age groups — Young Adults lead at 26.6% (\$62,143) while Seniors contribute 23.8% (\$55,763), indicating broad cross-generational appeal with **Young Adults as the top-revenue segment**.

Shipping Type Performance

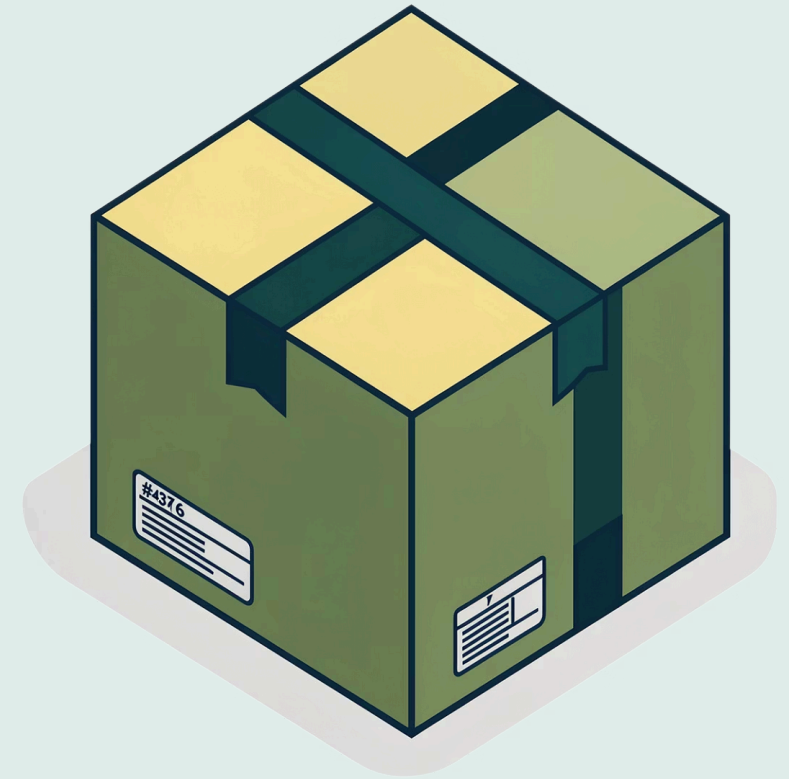
Standard Shipping

Avg purchase: **\$58.46**

Express Shipping

Avg purchase: **\$60.48**

Express users spend **3.5% more per transaction** — a clear signal to leverage premium shipping as part of an upsell and premium positioning strategy.



Key Insights & Patterns

1

Loyalty Dominance, Weak Acquisition

Loyal segment is **80%** of the base, but new customer acquisition sits at just **2%** — a pipeline risk.

2

Gender Revenue Gap

Male customers drive **68% of revenue**; the female segment remains underexploited.

3

Non-Subscriber Revenue

Non-subscribers generate **73% of revenue** despite lower subscription penetration.

4

Discount Margin Risk

Discount-dependent products — Hat, Sneakers, Coat — need an urgent **margin review**.

Strategic Recommendations



Boost Subscriptions

Promote exclusive benefits and perks to convert the large non-subscriber base into loyal members.



Strengthen Loyalty Programs

Reward repeat buyers to deepen the Loyal segment and reduce churn risk.



Review Discount Policy

Balance volume gains with margin protection — especially for Hat, Sneakers, and Coat.



Targeted Marketing

Focus on high-revenue age groups and express-shipping users for premium upsell opportunities.

